HEALTHCARE DATA ANALYTICS DASHBOARD - POWER BI

1. PROJECT OVERVIEW

The **Healthcare Data Analytics Dashboard** is an interactive data analytics project developed using **Microsoft Power BI** to analyze patient demographics, health indicators, hospital utilization, and treatment costs.

The project transforms raw patient-level healthcare data into meaningful business insights through data preparation, data modeling, DAX measures, KPI development, interactive filters, and visual analysis.

The dashboard enables users to explore healthcare data based on factors such as age, gender, diabetes status, smoking status, hospital visits, BMI, and treatment cost.

The primary objective of this project is to demonstrate how Power BI can be used to convert raw healthcare data into an interactive analytical solution that supports data-driven decision-making.

2. BUSINESS PROBLEM

Healthcare organizations generate large amounts of patient data containing demographic, health, utilization, and cost-related information.

However, raw datasets are difficult to analyze directly and may not provide an immediate understanding of important patterns such as:

- Patient demographic distribution
- Age-wise hospital utilization
- Diabetes and smoking prevalence
- Average BMI
- Hospital visit patterns
- Treatment cost levels
- Differences across patient groups

A centralized and interactive dashboard can help transform this raw information into an easily understandable analytical view.

Therefore, this project focuses on developing a Power BI dashboard that provides a consolidated view of patient demographics, health indicators, hospital utilization, and treatment costs.

3. BUSINESS OBJECTIVES

The key objectives of the project are:

- Analyze the demographic distribution of patients by age and gender.
- Segment patients into meaningful age groups.
- Analyze diabetes and smoking status across the patient population.
- Measure total and average hospital visits.
- Analyze average BMI across patients.
- Analyze average treatment cost.
- Explore BMI and treatment cost at the patient level.
- Compare healthcare indicators across different demographic groups.
- Provide interactive filtering using Power BI slicers.
- Present the findings through a clear and management-friendly dashboard.

4. BUSINESS QUESTIONS

The dashboard was designed to answer the following analytical questions:

Patient Demographics

a) How many patients are included in the dataset?
b) What is the distribution of patients by gender?
c) How are patients distributed across different age groups?
d) Which age group contains the highest number of patients?

Health Indicators

e) What proportion of patients are smokers?
f) What proportion of patients have diabetes?
g) What is the average BMI of the patient population?

Hospital Utilization

h) What is the total number of hospital visits?
i) What is the average number of hospital visits per patient?
j) How do average hospital visits vary across age groups?

Treatment Cost

k) What is the average treatment cost?
l) How does treatment cost vary across individual patients?
m) How can BMI and treatment cost be explored by gender?

Interactive Analysis

n) How do the key metrics change when filtering by diabetes status, gender, or smoking status?

5. DATASET OVERVIEW

The project uses a healthcare patient dataset containing **500 patient records** and **10 columns**.

Dataset Fields

Column	Description
Patient_ID	Unique identifier for each patient
Age	Age of the patient
Gender	Gender of the patient
BMI	Body Mass Index
Blood_Pressure	Blood pressure measurement
Cholesterol_Level	Cholesterol level
Diabetes	Diabetes status
Smoker	Smoking status
Hospital_Visits	Number of hospital visits
Treatment_Cost	Treatment cost associated with the patient

Dataset Size

- **Total Records:** 500
- **Total Columns:** 10
- **Age Range:** 18–90 years
- **Gender Categories:** Male, Female
- **Diabetes Categories:** Yes, No
- **Smoking Categories:** Yes, No

6. DATASET PREVIEW

The dataset contains patient-level information covering demographic characteristics, health indicators, hospital utilization, and treatment costs.

The dataset was subsequently prepared and analyzed in Power BI.

The following section shows a preview of the original healthcare dataset used for analysis.

Patient_ID	Age	Gender	BMI	Blood_Pressure	Cholesterol_Level	Diabetes	Smoker	Hospital_Visits	Treatment_Cost
P0001	69	Male	30.3	129	193	No	No	4	12004.33
P0002	32	Male	22.2	93	164	Yes	No	13	2621.49
P0003	89	Male	29.5	143	192	Yes	Yes	9	1817.13
P0004	78	Male	18.9	126	236	No	No	4	13670.47
P0005	38	Female	S	108	219	Yes	Yes	9	16642.88
P0006	41	Female	28.3	155	195	Yes	Yes	12	15858.29
P0007	20	Female	16.3	115	257	Yes	No	4	11332.31
P0008	39	Female	23.4	178	272	No	Yes	9	2897.47
P0009	70	Male	18.4	131	170	Yes	Yes	11	14072.84
P0010	19	Male	16.6	155	300	No	No	4	5595.89
P0011	47	Male	39.7	86	256	Yes	No	2	18310.31
P0012	55	Female	23.1	108	229	No	Yes	6	16927.04
P0013	19	Female	35.2	90	196	Yes	Yes	5	14478.28
P0014	81	Male	21.4	112	266	No	No	3	19028.07
P0015	77	Male	32	85	273	No	Yes	1	1597.01
P0016	38	Female	34	158	292	Yes	Yes	11	14987.81
P0017	50	Female	29.9	111	174	No	Yes	8	7274.61
P0018	75	Male	26.8	130	185	No	Yes	0	18215.25
P0019	39	Male	25.3	154	200	No	Yes	12	6066.83
P0020	66	Female	23.7	137	164	No	Yes	12	14772.27

7. DATA CLEANING AND TRANSFORMATION

Before developing the dashboard, the dataset was reviewed and prepared for analysis.
The following data preparation activities were performed:

7.1 Data Type Validation

The columns were checked and assigned appropriate data types:

- Patient_ID → Identifier
- Age → Whole Number
- Gender → Text
- BMI → Decimal Number
- Blood_Pressure → Numeric
- Cholesterol_Level → Numeric
- Diabetes → Text
- Smoker → Text
- Hospital_Visits → Whole Number
- Treatment_Cost → Decimal Number

7.2 Data Quality Checks

The dataset was reviewed for:

- Missing values
- Duplicate patient records
- Incorrect data types
- Invalid age values
- Invalid categorical values
- Unexpected numerical ranges

7.3 Age Group Creation

Age groups were created to make age-based analysis easier.

The recommended age-group structure is:

- **18–30**
- **31–45**
- **46–60**
- **61+**

This grouping provides a clearer comparison of hospital utilization across different stages of the patient population.

8. DATA MODELING

The project uses the healthcare dataset as the primary analytical table.
The data model was designed to support:

- Patient-level analysis
- Demographic analysis
- Health indicator analysis
- Hospital utilization analysis
- Treatment cost analysis
- Interactive filtering

The main analytical fields include:

Patient_ID, Age, Gender, BMI, Diabetes, Smoker, Hospital_Visits, and Treatment_Cost.
The model supports aggregation through Power BI measures and filtering through slicers.

9. DAX MEASURES

The following DAX measures can be used to calculate the major KPIs in the dashboard.

Total Patients
Total Patients =
DISTINCTCOUNT(Healthcare[Patient_ID])

Total Hospital Visits
Total Hospital Visits =
SUM(Healthcare[Hospital_Visits])

Average Hospital Visits
Average Hospital Visits =
AVERAGE(Healthcare[Hospital_Visits])

Average BMI
Average BMI =
AVERAGE(Healthcare[BMI])

Average Treatment Cost
Average Treatment Cost =
AVERAGE(Healthcare[Treatment_Cost])

Smoker Patients
Smoker Patients =
CALCULATE(
DISTINCTCOUNT(Healthcare[Patient_ID]),
Healthcare[Smoker] = "Yes"
)

Diabetes Patients

```
Diabetes Patients =
CALCULATE(
DISTINCTCOUNT(Healthcare[Patient_ID]),
Healthcare[Diabetes] = "Yes"
)
```

These measures provide reusable calculations that can respond dynamically to dashboard filters.

10. POWER BI DASHBOARD OVERVIEW

The dashboard was designed as an interactive healthcare analytics report.

The dashboard contains the following analytical sections:

Patient Demographics

- Age Group Distribution
- Age Distribution
- Gender Distribution

Health Indicators

- Smoker Portion
- Diabetes Portion
- Average BMI

Hospital Utilization

- Total Hospital Visits
- Average Hospital Visits by Age Group

Treatment Cost

- Average Treatment Cost
- Treatment Cost by Patient
- BMI by Patient

Interactive Filters

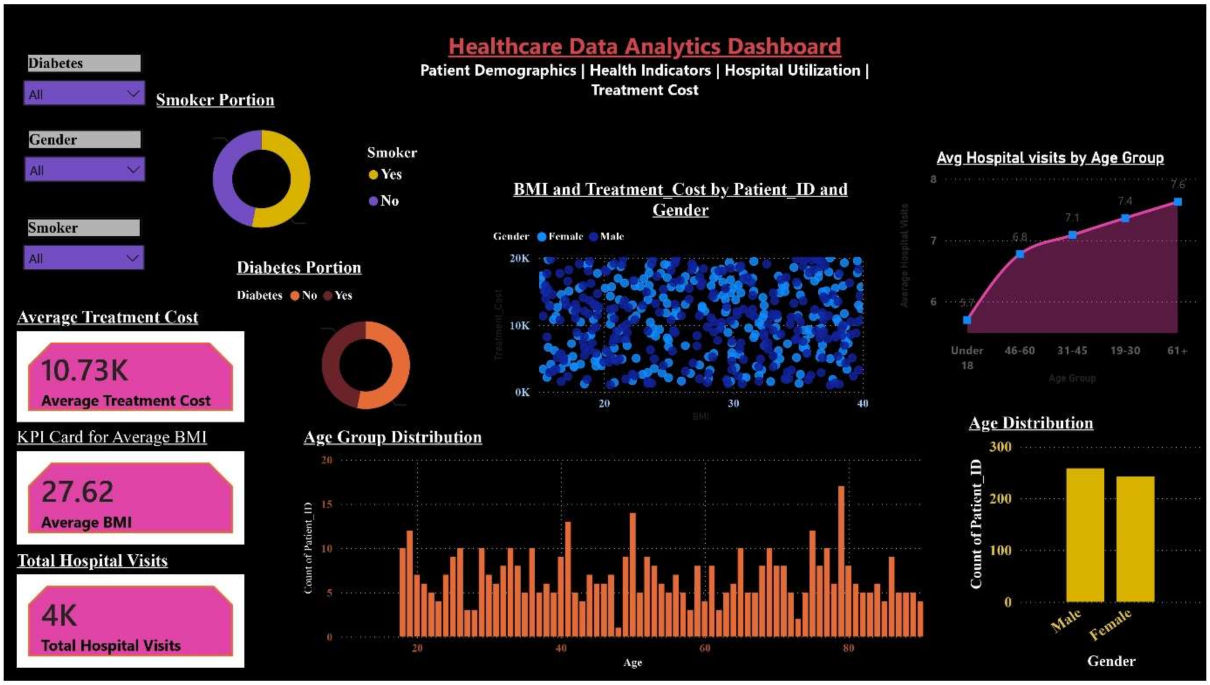
The dashboard includes slicers for:

- Diabetes
- Gender
- Smoker

These filters allow users to dynamically explore the dataset.

11. <u>POWER BI DASHBOARD SCREENSHOT</u>

Dashboard: Healthcare Data Analytics Dashboard



12. <u>KEY PERFORMANCE INDICATORS</u>

The main KPIs identified from the dataset are:

KPI	Value
Total Patients	500
Total Hospital Visits	3,631
Average Hospital Visits	7.26
Average BMI	27.62
Average Treatment Cost	10,731.44

The dashboard may display some KPI values in rounded/abbreviated form. For example, total hospital visits may appear as approximately **4K** and average treatment cost as approximately **10.73K** depending on Power BI formatting.

13. <u>DEMOGRAPHIC ANALYSIS</u>

13.1 Gender Distribution

The dataset contains:

- **Male:** 258 patients
- **Female:** 242 patients

The patient population is relatively balanced by gender, with a slightly higher number of male patients.
This provides a reasonable basis for comparing healthcare indicators across gender categories.

13.2 Age Distribution

The dataset contains patients between **18 and 90 years of age**.

The recommended age groups are:

Age Group	Patients
18–30	93
31–45	107
46–60	97
61+	203

The **61+ age group contains the largest number of patients**, representing 203 of the 500 records.
This makes the 61+ group an important segment for further utilization and cost analysis.

14. HEALTH INDICATOR ANALYSIS

14.1 Smoking Status

The dataset contains:

- **Smokers:** 266 patients
- **Non-smokers:** 234 patients

Therefore:

- Smokers = **53.2%**
- Non-smokers = **46.8%**

The dashboard provides an interactive view of smoking status and allows users to filter other healthcare metrics based on smoker status.

14.2 Diabetes Status

The dataset contains:

- **Diabetes: Yes:** 234 patients
- **Diabetes: No:** 266 patients

Therefore:

- Diabetes = **46.8%**
- No Diabetes = **53.2%**

The dashboard allows diabetes status to be used as an interactive filter for further analysis.

14.3 Average BMI

The average BMI of the patient population is:**27.62**
BMI is included as one of the primary health indicators in the dashboard.
The dashboard also allows BMI values to be explored at the patient level.

15. HOSPITAL UTILIZATION ANALYSIS

15.1 Total Hospital Visits

The dataset contains a total of: **3,631 hospital visits**
This represents the total number of recorded hospital visits across all 500 patients.

15.2 Average Hospital Visits

The average number of hospital visits per patient is: **7.26 visits**
This KPI provides an overall measure of hospital utilization across the patient population.

15.3 Hospital Visits by Age Group

The average hospital visits by age group are:

Age Group	Average Hospital Visits
18–30	7.36
31–45	7.08
46–60	6.77
61+	7.63

The **61+ age group has the highest average hospital visits at approximately 7.63 visits per patient**.
The 46–60 age group has the lowest average at approximately 6.77 visits.
These findings indicate differences in hospital utilization across age groups within this dataset.

16. TREATMENT COST ANALYSIS

16.1 Average Treatment Cost

The average treatment cost is: **10,731.44**
The dashboard displays this KPI in an abbreviated format of approximately: **10.73K**

16.2 Treatment Cost by Patient

The dashboard includes a patient-level view of treatment cost.
This allows users to identify differences in treatment costs across individual patients.
Treatment cost can also be explored alongside gender and BMI to support more detailed analysis.

17. BMI AND TREATMENT COST ANALYSIS

The dashboard includes patient-level visualizations for:

- BMI
- Treatment Cost
- Gender

These visualizations allow users to examine how patient-level BMI and treatment cost values are distributed across the dataset.

The dashboard should be interpreted as an analytical exploration rather than evidence of a direct causal relationship between BMI and treatment cost.

18. INTERACTIVE DASHBOARD FEATURES

The Power BI dashboard includes interactive slicers for:

Diabetes

Users can filter the dashboard based on:

- Diabetes = Yes
- Diabetes = No

Gender
Users can filter the dashboard based on:

- Male
- Female

Smoker
Users can filter the dashboard based on:

- Smoker = Yes
- Smoker = No

These filters allow users to perform dynamic comparisons without manually changing individual charts.

19. KEY INSIGHTS

Based on the analysis, the following insights were identified:

Insight 1 – Older Patients Represent the Largest Segment

The **61+ age group contains 203 patients**, making it the largest age segment in the dataset.

Insight 2 – Hospital Utilization Is Highest Among the 61+ Group

The 61+ group has the highest average hospital visits at approximately **7.63 visits per patient**.

Insight 3 – Gender Distribution Is Relatively Balanced

The dataset contains **258 male patients and 242 female patients**, resulting in a relatively balanced gender distribution.

Insight 4 – Smoking Status Is Slightly Higher for Smokers

There are **266 smokers compared with 234 non-smokers**, meaning smokers represent approximately **53.2%** of the dataset.

Insight 5 – Diabetes Status Is Slightly More Common in the Non-Diabetes Group

There are **234 patients with diabetes and 266 patients without diabetes**.
Therefore, diabetes represents approximately **46.8%** of the dataset.

Insight 6 – Average BMI Is 27.62

The overall average BMI is **27.62**, providing a high-level view of the BMI distribution of the patient population.

Insight 7 – Average Treatment Cost Is Approximately 10.73K

The average treatment cost is **10,731.44**, which can be displayed as approximately **10.73K** in the Power BI dashboard.

20. BUSINESS RECOMMENDATIONS

Based on the dashboard analysis, the following analytical recommendations can be considered:

20.1 Focus on the Older Patient Segment

Since the 61+ age group represents the largest patient segment and has the highest average hospital visits, healthcare teams can prioritize this segment for deeper utilization analysis.

20.2 Monitor Hospital Utilization

Hospital visit patterns should be monitored across age groups to identify changes in healthcare utilization over time.

20.3 Segment Analysis Using Interactive Filters

Healthcare analysts can use diabetes, gender, and smoker filters to perform targeted comparisons across patient segments.

20.4 Monitor Treatment Cost Patterns

Treatment costs can be analyzed at the patient and demographic levels to identify unusually high or low cost patterns requiring further investigation.

20.5 Extend the Analysis

Future versions of the dashboard could analyze relationships between:

- Diabetes and hospital visits
- Smoking and hospital visits
- Age and treatment cost
- BMI and treatment cost
- Cholesterol and treatment cost
- Blood pressure and hospital utilization

These analyses should be treated as exploratory relationships and should not automatically be interpreted as causal relationships.

21. PROJECT LIMITATIONS

The project has several limitations:

1. The dataset contains only 500 patient records.
2. The dataset does not provide a time dimension for trend analysis.
3. The project does not establish medical or clinical causality.
4. Treatment cost relationships require additional variables for deeper analysis.
5. The dataset may not represent the broader healthcare population.
6. Additional healthcare attributes could provide more detailed predictive analysis.

22. FUTURE ENHANCEMENTS

The project can be extended by adding:

Advanced Analytics

- Predictive modeling
- Patient risk scoring
- Hospital visit prediction
- Treatment cost prediction

Additional Data

- Diagnosis information
- Medical procedures
- Medication information
- Insurance information
- Admission and discharge dates
- Hospital/department information

Advanced Power BI Features

- Drill-through pages
- Tooltips
- Bookmark navigation
- Time-series analysis
- Dynamic KPI cards
- Conditional formatting
- Advanced segmentation

23. TOOLS AND TECHNOLOGIES

Tools Used

- Microsoft Power BI
- Power Query
- DAX
- Microsoft Excel / CSV

Skills Demonstrated

- Data Cleaning
- Data Transformation
- Data Modeling
- Exploratory Data Analysis
- KPI Development
- DAX
- Data Visualization
- Dashboard Development
- Business Insight Generation
- Interactive Reporting

24. PROJECT WORKFLOW

The project followed the following analytical workflow:

Raw Healthcare Dataset
↓
Data Validation & Cleaning
↓
Data Transformation
↓
Age Group Creation
↓
Data Modeling
↓
DAX Measures
↓
Exploratory Analysis
↓
Power BI Visualizations
↓
Interactive Dashboard
↓
Insights & Recommendations

25. CONCLUSION

The Healthcare Data Analytics Dashboard demonstrates how Power BI can transform raw patient-level healthcare data into an interactive and business-oriented analytical solution.
The project provides insights into patient demographics, health indicators, hospital utilization, and treatment costs.

The analysis identified that the **61+ age group is the largest patient segment and has the highest average hospital utilization**, while the overall patient population has an average BMI of **27.62** and an average treatment cost of approximately **10.73K**.
Interactive slicers for diabetes, gender, and smoking status allow users to perform deeper segment-level analysis.

Overall, the project demonstrates practical skills in **data preparation, Power BI dashboard development, DAX, data visualization, analytical thinking, and business insight generation**.